Response to Reviewers

Resubmission of T-SP-35123-2026

We thank the Associate Editor and the three reviewers for their detailed and constructive evaluations. This is a substantially revised resubmission. In response to the central concern shared by all reviewers—that the original manuscript read as an *integration* of existing methods—we have **reframed the paper around the closed-loop coupling itself as a front-end-agnostic principle**, and made the contributions **theoretical**: a posterior Cramér–Rao bound for the complete closed loop (Theorem 1) and a minimum-variance guarantee for the tracker-aided subspace refinement (Theorem 2). We have also **corrected every flagged derivation**, replaced the unsupported "exact" CRLB and the heuristic convergence bound, and **re-run all experiments with up to** 500 **Monte-Carlo trials and** 95% **confidence intervals**, adding a closed-loop-MUSIC baseline, a 2×2 (front-end $\times$ loop) study, and a back-end ablation against the LMB and δ-GLMB labeled-RFS filters. Crucially, we now **delineate the operating regime** in which the higher-order front-end helps rather than claiming uniform superiority. Concretely, the revised experiments **quantify the closed loop's efficacy** (consolidated in a new efficacy table): it restores identifiability where the data-only bound diverges (Theorem 1), lowers COP localization error by 56% (MUSIC's by 44%) at low SNR, and reduces moving-target identity switches by 15% at matched detection, while a validation gate and a motion-compensated prior guarantee it *never degrades* the open-loop baseline.

Summary of major changes.

1. **New title and framing**: "A Closed-Loop Subspace–RFS Framework for DOA Estimation and Multi-Target Tracking." The closed-loop refinement is shown to be front-end-agnostic and is instantiated with both COP (cumulant) and MUSIC (covariance) front-ends.

2. **Theorem 1 (closed-loop posterior CRB)** via the Tichavský–Muravchik–Nehorai recursion: proves the temporal prior strictly tightens the bound and restores identifiability where the data-only bound diverges. Replaces the former "exact" static cumulant CRLB.

3. **Theorem 2 (minimum-variance refinement)**: recasts T-COP refinement as a basis-invariant inverse-variance fusion on the Grassmann manifold with an explicit prior-accuracy condition, enforced by a validation gate.

4. **Corrected derivations**: effective sample size (Kish), velocity gating (no double counting), cumulant-model assumptions, and the cumulant-noise covariance ($\mathbf{\Sigma}_c \neq \sigma_c^2\mathbf{I}$).

5. **Honest re-evaluation**: Monte-Carlo (up to 500 trials) with CIs; GOSPA decomposition and matched-pair localization RMSE; MUSIC and closed-loop-MUSIC baselines; explicit regime delineation.

6. **Back-end renamed** to track-oriented PHD (TO-PHD) with an honest statement of its relation to the exact GM-PHD recursion.

7. **Back-end-agnostic with physics-based motion, plus robustness sweeps**: the framework drives LMB and full δ-GLMB back-ends, and a physics-based constant-acceleration δ-GLMB is best for maneuvering targets (our recommended deployed configuration). New sweeps confirm the gains hold across source count K (the determined$\to$underdetermined transition at $M-1$) and across non-Gaussian source signal types, with the expected higher-order-statistics limit for purely Gaussian sources.

Reviewer 1

R1.1 The theoretical concepts employed are all separately well-established; the manuscript does not clearly separate what is fundamentally new.

Response: We agree this was the core weakness. We have reframed the paper so the contribution is the *closed-loop coupling*, now backed by two theorems (Sec. V) that did not exist in the prior version. We further show the closed loop is front-end-agnostic (it improves both COP and MUSIC), which establishes it as a general principle rather than a specific combination. Contributions are now stated as two theory results plus one practical capability.

R1.2 Several mechanisms (T-COP, SD-COP, COP-spectrum birth, velocity-gated merging) appear heuristic or only lightly justified; present derivations/proofs more completely.

Response: T-COP is now derived as the BLUE on the Grassmann tangent space with a proof of minimum variance and an explicit benefit condition (Theorem 2). The velocity-gated merge is corrected (Eq. 25) and its role is tied to Theorem 2's gate. The COP-spectrum birth is *downgraded to an empirical detection score*; we no longer claim Neyman–Pearson optimality. SD-COP is explicitly labeled a *supporting* extension (Sec. III-C).

R1.3 The CRLB section does not develop a new CRLB for the complete closed-loop COP-RFS tracking framework.

Response: Addressed directly by Theorem 1, the posterior CRB for the coupled estimator–tracker, validated against Monte-Carlo simulation (Sec. VI).

R1.4 Compare different state-of-the-art tracking approaches.

Response: We added a *closed-loop MUSIC* baseline, a 2×2 (front-end $\times$ loop) ablation, and a *tracking back-end ablation against the labeled multi-Bernoulli (LMB) and full δ-GLMB filters*—state-of-the-art identity-aware RFS trackers—in addition to MUSIC, Capon, ESPRIT, and the MUSIC–PHD cascade (Sec. VI). All three labeled trackers (our TO-PHD, LMB, and δ-GLMB) cut identity switches by an order of magnitude over the standard GM-PHD, with the full δ-GLMB and LMB the most accurate, which both fairly benchmarks our back-end and shows the framework is back-end-agnostic. Sparse/SBL DOA methods are cited and discussed as related work.

R1.5 The reference list is somewhat limited for a full TSP paper.

Response: We broadened the references—posterior CRB (Tichavský *et al.*, 1998), SBL for DOA (Wipf–Rao 2004; Gerstoft *et al.* 2016), labeled-RFS/δ-GLMB (Vo–Vo 2013), and Grassmann geometry (Edelman *et al.* 1998)—each with a DOI in the manuscript bibliography.

R1.6 Significantly overlength; more suitable as a focused letter, or a good TSP paper with strengthened theory and novelty positioning.

Response: We chose the latter. The paper is reorganized around the two theorems and the underdetermined capability; lightly-justified mechanisms and redundant experiments were compressed, and the novelty positioning is rewritten (Abstract, Sec. I). The manuscript is now $\sim$10 pages (two-column IEEE format).

Reviewer 2

R2.1 Novelty lies mainly at the level of algorithmic integration; isolate and validate the individual contributions of T-COP, SD-COP, the modified tracker, and the closed-loop architecture.

Response: The closed-loop architecture is now the central, theoretically characterized contribution (Theorems 1–2). We added a 2×2 (front-end $\times$ loop) ablation that isolates the closed-loop effect from the front-end effect, showing the loop helps *both* COP and MUSIC, while the COP front-end's distinct value is confined to the underdetermined regime.

R2.2 T-COP subspace interpolation (Eq. 13) is basis-dependent; sign/phase/ unitary transformations give different refined subspaces.

Response: Correct. Eq. (13) is replaced by an orthogonal-Procrustes alignment followed by an inverse-variance fusion reported through the projector $\mathbf{P}_s = \mathbf{U}_s\mathbf{U}_s^H$, which is invariant to the basis gauge $\mathbf{U}_s \mapsto \mathbf{U}_s\mathbf{Q}$ (Sec. III-B, Theorem 2).

R2.3 SD-COP's super-underdetermined claim is supported mainly by simulation.

Response: SD-COP is now a supporting extension, connected to the corollary of Theorem 1: sources beyond the static virtual-array limit remain identifiable once the temporal prior makes the predicted information positive definite on the data-unresolved subspace.

R2.4 Fig. 12 caption says "tracks all targets" while the text says "majority"; $P_d = 60\%$ vs 51% is a limited improvement without confidence intervals; Table I shows MUSIC+Physics beating COP+Physics; avoid "substantially outperforms"/"tracks all".

Response: All overclaims removed. We now report means with 95% CIs over up to 500 trials. The underdetermined detection advantage is $P_d = 60.3\% \pm 0.2$ vs. $51.9\% \pm 0.1$ (non-overlapping CIs). We explicitly state that in the *determined* regime MUSIC is superior (lower GOSPA and localization error), and we frame COP's advantage as specific to $K > M - 1$. "substantially outperforms"/"tracks all" are deleted throughout.

R2.5 Add stronger baselines: SBL/sparse reconstruction for DOA; LMB/ δ-GLMB for identity-aware tracking. Increase Monte-Carlo trials.

Response: Key experiments now use up to 500 Monte-Carlo trials with 95% CIs (up to a $50\times$ increase over the prior 10). *Both* requested identity-aware trackers are now implemented and compared: we benchmark the proposed track-oriented PHD against the standard GM-PHD, the *labeled multi-Bernoulli (LMB) filter*, and the *full δ-GLMB filter* (Vo–Vo 2013; Gibbs implementation of Vo–Vo–Hoang 2017) (Sec. VI back-end ablation, Table on crossing targets): all three labeled trackers reduce identity switches by an order of magnitude over the GM-PHD, with the full δ-GLMB giving the cleanest identity (fewest switches) and sharpest localization, LMB the best aggregate GOSPA, and the TO-PHD a competitive lightweight alternative—fairly benchmarking our back-end against the state of the art. A closed-loop-MUSIC front-end is also added. Sparse/SBL DOA methods are cited and discussed as related work; we focus the quantitative front-end comparison on the most directly comparable subspace estimators (MUSIC, closed-loop MUSIC).

Reviewer 3

R3.1 The derivation from the raw cumulant tensor to $\mathbf{C}_{2\rho} = \mathbf{A}_v\mathbf{\Gamma}_s\mathbf{A}_v^H$ is incomplete / oversimplified.

Response: We now state the conditions (A1 independence, A2 non-Gaussianity, A3 nonzero 2ρ-order autocumulants) under which the tensor reduces to the Hermitian Toeplitz model with full-column-rank $\mathbf{A}_v$, with citations to the established construction (Sec. II-B).

R3.2 Eq. (5) assumes a diagonal $\mathbf{\Gamma}_s$ requiring independent non-Gaussian sources; for Gaussian/weakly non-Gaussian sources the gain vanishes.

Response: Stated explicitly: under (A1)–(A3) cross-cumulants vanish giving the diagonal $\mathbf{\Gamma}_s$, and for Gaussian/vanishing-cumulant sources the gain degenerates to the classical $M-1$ limit (Sec. II-B).

R3.3 Effective sample size should scale as $(\sum w_i)^2/\sum w_i^2$, not $T(1-\alpha^n)/(1-\alpha)$; Eq. (10) overestimates it.

Response: Corrected to the Kish effective sample size, $T_{\text{eff}} = (\sum w_i)^2/\sum w_i^2 \to T(1+\alpha)/(1-\alpha)$ (Eq. 10). We note the corrected expression and that the naive geometric sum is not the effective sample size.

R3.4 Eq. (13) linear interpolation of subspace bases is basis-dependent and not a principled Grassmann interpolation.

Response: Addressed identically to R2.2: Procrustes alignment + inverse-variance fusion on the Grassmann tangent space, proven minimum-variance (Theorem 2).

R3.5 Replacing the PHD update by Hungarian + single-measurement Kalman changes the recursion without a new posterior-intensity derivation; it is closer to heuristic track-before-update association.

Response: We agree and now *name it accordingly*: a track-oriented PHD (TO-PHD). A new paragraph (Sec. IV) states precisely that the matched single-measurement update is a track-oriented, marginal-association approximation (the PHD analogue of MHT/GNN), not the exact GM-PHD corrector, and that it reduces to the GM-PHD when the gate is fully opened.

R3.6 Eq. (25) uses the full-state difference with $\mathbf{P}_i^{-1}$ and then gates velocity separately, double-counting velocity.

Response: Corrected: we use the position sub-block $(\theta_i - \theta_j)^2/[\mathbf{P}_i]_{11}$ and gate velocity separately, with an explicit note that a full-state Mahalanobis term would double-count the velocity (Eq. 25).

R3.7 The "exact" higher-order CRLB substitutes a virtual steering matrix into a covariance-based form; cumulant-domain noise is not white ($\sigma_c^2\mathbf{I}$).

Response: The "exact" claim is removed. We present the static cumulant CRLB as an approximate white-noise surrogate, give the general Slepian–Bangs form with the non-white $\mathbf{\Sigma}_c$, and provide the rigorous closed-loop bound in Theorem 1.

R3.8 Proposition 1 lacks treatment of missed detections/clutter/multiple targets; Proposition 2's Neyman–Pearson "sufficient statistic" claim has no probability model; Theorem 1's convergence bound is heuristic.

Response: Proposition 1 now includes the detection/clutter factor $p_D^2(1-p_{\text{fa}})$. Proposition 2 is removed; COP-spectrum birth is now presented as a heuristic detection score without an optimality claim. The former heuristic convergence "Theorem 1" is removed and replaced by the contraction-to-PCRB-floor statement implied by Theorems 1–2.

R3.9 Very few Monte-Carlo trials (e.g., 10); "tracks all" vs. $\sim 60\%$ detection is internally inconsistent.

Response: All key results now use ≥ 500 trials with 95% CIs, and the inconsistency is removed (Sec. VI).

R3.10 Several components are inherited or used heuristically; the closed-loop theoretical contribution is not convincingly established.

Response: The closed-loop contribution is now established by Theorems 1–2 and an ablation isolating it from the front-end, and validated against simulation.